

Namhyuk Ahn

ASSISTANT PROFESSOR AT INHA UNIVERSITY

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Research Interests

My research vision is to establish a comprehensive **Generative AI + X ecosystem** that empowers a diverse spectrum of users, from domain experts to the general public. I strive to democratize the benefits of AI by extending them to under-resourced fields, thereby maximizing the societal utility of generative technologies. My work centers on three key pillars: (1) **AX via Gen AI**: Engineering domain-specific adaptations of foundation models for specialized sectors that necessitate high-fidelity, bespoke solutions, (2) **Professional-Grade Gen AI Systems**: Designing workflows that satisfy rigorous professional requirements through fine-grained controllability and steerability, and (3) **Responsible Gen AI**: Developing robust frameworks for data security, privacy protection, and ethical alignment to safeguard all stakeholders.

Work Experience

Inha University

ASSISTANT PROFESSOR

- School of Electrical and Electronic Engineering
- Department of Electrical and Computer Engineering (Graduate School)

Incheon, Korea

Sep. 2024 - Present

NAVER WEBTOON

AI RESEARCHER

- Built user-centric AI tools designed for cartoon creators/artists (e.g. Cartooner, DreamStyler)
- Researched user privacy-aware Gen AI (e.g. Impasto, which prevents copyright violation by Gen AI models)
- Developed portrait stylization system (e.g. WebtoonMe)

Pangyo, Korea

Aug. 2021 - Aug. 2024

NAVER AI LAB

VISITING RESEARCHER

- Researched data augmentation for image super-resolution (e.g. CutBlur)
- Developed label-efficient conditional generative models
- Co-worked with Jaejun Yoo, Youngjung Uh and Yunjey Choi

Bundang, Korea

Sep. 2019 - Oct. 2020

NAVER

INTERN

- Developed image-to-image translation pipeline for talking head project

Bundang, Korea

June 2018 - Aug. 2018

Education

Ajou University

PH.D. IN ARTIFICIAL INTELLIGENCE

- Advisor: Prof. Kyung-Ah Sohn
- Thesis: Toward an Efficient Deep Image Restoration Method

Suwon, Korea

Mar. 2016 - Aug. 2021

Ajou University

BACHELOR OF MEDIA IN DIGITAL MEDIA

Suwon, Korea

Mar. 2012 - Aug. 2016

Selected Publications

- Minsuk Ji*, Sanghyeok Lee*, and **Namhyuk Ahn**. Compositional Image Synthesis with Inference-Time Scaling. **ICASSP 2026**
- Soyoung Yoon, **Namhyuk Ahn**, and In Kyu Park. T2LF: LLM-Guided Multimodal Diffusion for Text-to-Light Field Synthesis. **WACV 2026**
- Sungnyun Kim, Junsoo Lee, Kibeom Hong, Daesik Kim, and **Namhyuk Ahn**. DiffBlender: Scalable and Composable Multimodal Text-to-Image Diffusion Models. **Expert Systems with Applications (ESWA) 2026**
- **Namhyuk Ahn**, Wonhyuk Ahn, KiYoon Yoo, Daesik Kim, and Seung-Hun Nam. Imperceptible Protection Against Style Imitation from Diffusion Models. **IEEE Transactions on Multimedia (TMM) 2026**

- **Namhyuk Ahn**, KiYoon Yoo, Wonhyuk Ahn, Daesik Kim, and Seung-Hun Nam. Nearly Zero-Cost Protection Against Mimicry by Personalized Diffusion Models. **CVPR 2025**
- Jihye Back, **Namhyuk Ahn**, and Jangho Kim. Magnitude Attention-based Dynamic Pruning. **Expert Systems with Applications (ESWA) 2025**
- **Namhyuk Ahn**, Junsoo Lee, Chunggi Lee, Kunhee Kim, Daesik Kim, Seung-Hun Nam, and Kibeom Hong. DreamStyler: Paint by Style Inversion with Text-to-Image Diffusion Models. **AAAI 2024**
- **Namhyuk Ahn**, Jaejun Yoo, and Kyung-Ah Sohn. Data Augmentation for Low-Level Vision: CutBlur and Mixture-of-Augmentation. **International Journal of Computer Vision (IJCV) 2024**
- Jeong-Hyeon Moon*, **Namhyuk Ahn***, and Kyung-Ah Sohn. Decomposing Texture and Semantics for Out-of-distribution Detection. **Expert Systems with Applications (ESWA), 2024**
- **Namhyuk Ahn**, Patrick Kwon, Jihye Back, Kibeom Hong, and Seungkwon Kim. Interactive Cartoonization with Controllable Perceptual Factors. **CVPR 2023**
- **Namhyuk Ahn**, Byungkong Kang, and Kyung-Ah Sohn. Efficient Deep Neural Network for Photo-Realistic Image Super-Resolution. **Pattern Recognition (PR) 2022**
- Jaejun Yoo*, **Namhyuk Ahn***, and Kyung-Ah Sohn. Rethinking Data Augmentation for Image Super-Resolution: A Comprehensive Analysis and a New Strategy. **CVPR 2020**
- **Namhyuk Ahn**, Byungkong Kang, and Kyung-Ah Sohn. Fast, Accurate, and Lightweight Super-Resolution With Cascading Residual Network. **ECCV 2018**

Professional Service

Reviewer CVPR, ICCV, ECCV, NeurIPS, ICLR, ICML, AAAI, ACMMM, ARR, WACV
TPAMI, IJCV, TIP, TMM, TCSVT, SPIC, ESWA, Neurocomputing

PC IPIU (2027)

Teaching

Undergraduate Deep Learning (2025-), Multimedia (2025-), Python Programming (2025-),
Computer Graphics (2024), Creative Design for Engineering (2024)

Graduate Large Language Models (2026), Vision-Language Models (2025), Multimodal AI (2025)

Others Deep Learning and its Applications, FastCampus (2017)

Awards

2024.1 **NAVER N Innovation Award**, 3rd prize on R&D track

2023.1 **NAVER N Innovation Award**, 2nd prize on R&D track

2018.6 **Honorable Mention Award**, SR Challenge on NTIRE workshops @ CVPR

Invited Talks

2026.2 **IPIU**, Securing Visual Copyright: Recent Advances in Image Protection Against Diffusion Models

2025.10 **ICCE-Asia (Tutorial)**, Responsible Generative AI: From Provenance to Protection

2025.9 **SNU Hospital**, Useful or Harmful? Rethinking Generative AI in Vision

2025.7 **KCC**, Content Protection in Generative AI Era

2023.3 **Ajou University**, Recent Advances in Visual Generative Models

Press Mentions

2024.4 **Digital daily**, “네이버웹툰, ‘작품 무단 학습 AI’ 막는 모델 선보인다”

2023.2 **Tech M**, “미국·유럽·일본도 ‘감탄’... 웹툰 AI 시대 열어야죠”

2023.1 **Digital daily**, “내 얼굴이 웹툰 캐릭터로 변했다”